

Supporting Material: Artifact-Level Detail for a Condition-Indexed Operating Envelope and Representation-Repair Study in Automatic Modulation Classification

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Abstract

This is optional supporting material. It is written to be read on its own; the manuscript it accompanies states every result it relies on and points here only for audit-level tables. The purpose is narrower: to expose the artifact-level detail behind those results, so that a reviewer who wants to audit the construction of the evaluation cells, the full capacity-control sweep, the per-SNR comparator-fidelity curves, the covariate ladder and selector audits, the leave—15-dB-out extrapolation and paired overlap-increment tests, the comparator convergence histories, the sidecar architecture, or the seed-level robustness can do so directly. It also reports two preregistered post-hoc robustness validations on an independently generated severe held set: a transfer check that falsifies the severe-corner advantage and the envelope’s —15-dB transport, and a repair check showing that the received-I/Q sidecar restores the compact model on the same held set. Nothing in this document introduces a new claim, a new experiment, or a new evidence class.

I. SCOPE AND HOW TO READ THIS DOCUMENT

The study behind this material asks when a compact spectral classifier and a much larger I/Q-domain classifier exchange rank under structured interference, describes that exchange with a two-variable empirical envelope, and then diagnoses and repairs the representation boundary the exchange reveals. All evidence is from source-disjoint 3GPP TR 38.901 TDL-profile simulation.

Because this document is standalone, the identifiers used throughout are restated here.

- **A0** is the direct spectral reference backbone (8,458 parameters) and **A5** is the sealed complete configuration (39,500 parameters), which adds a three-route spectral allocation, a simulation-component teacher, multi-task and contrastive objectives, a residual bypass, and a small route-orthogonality regularizer.
- **S**, **M**, and **L** are prospective spectral capacity tiers. *S* is the sealed A5 configuration; **M** and **L** widen only the spectral-channel, embedding, and environment dimensions, to 99,596 and 233,244 parameters.
- **sidecar** is a prospective received-I/Q repair branch that raises the total model size from 39,500 to 46,794 parameters.
- **IQFormer-inspired** and **MCLDNN** are architecture-faithful unified-retraining references, not exact reproductions of every original training recipe.
- The **target-support jammer overlap** o is the fraction of the jammer’s total STFT power that falls inside the target’s 99%-energy time–frequency support. It is not an occupied-bandwidth fraction.

Two evidence classes are kept apart everywhere. The *sealed* family is the original frozen confirmatory set; everything else, including the envelope, the severe-interference stratification, the Tier-2 interventions, and the capacity ladder, is *exploratory or post-hoc* and cannot change a sealed conclusion.

TABLE I
CELL CONSTRUCTION FOR THE OPERATING ENVELOPE. “WINDOWS PER CELL” IS THE RANGE OF SOURCE-DISJOINT TEST WINDOWS CONTRIBUTING TO A CELL.

Split	Role	SIR strata (dB)	Cells	Windows per cell
id_test	fit	-10, -5, 0, +5, +10	16	230–1053
hard_interference	fit	-15, -10, -5, 0	16	290–343
unseen_jammer	held	-10, -5, 0, +5, +10	20	229–276
unseen_speed	held	-10, -5, 0, +5, +10	20	228–277
heldout_channel	held	-10, -5, 0, +5, +10	20	216–279
combined_ood	held	-10, -5, 0, +5, +10	20	223–270
clean_retention	excluded		0	1
Fit total			32	
Primary held total			80	

TABLE II
HELD-REGIME ENVELOPE SKILL. RMSE VALUES IN PERCENTAGE POINTS.

Reference	Held regime	Cells	Env. Const.	R^2_{skill}	Oracle	Sign
IQFormer-insp.	unseen speed	20	3.94	7.87	0.75	6.71 1.000
IQFormer-insp.	held-out channel	20	4.86	7.59	0.59	6.95 1.000
IQFormer-insp.	combined OOD	20	7.50	9.23	0.34	4.24 0.950
IQFormer-insp.	unseen jammer	20	7.74	8.67	0.20	3.83 0.950
IQFormer-insp.	pooled (80)	80	6.23	8.37	0.45	6.04 0.975
MCLDNN	unseen speed	20	4.66	8.88	0.72	7.78 1.000
MCLDNN	held-out channel	20	5.41	9.06	0.64	8.41 1.000
MCLDNN	combined OOD	20	10.23	10.28	0.01	2.71 0.900
MCLDNN	unseen jammer	20	10.34	10.24	-0.02	3.23 0.900
MCLDNN	pooled (80)	80	8.10	9.63	0.29	6.81 0.950

II. CONSTRUCTION OF THE EVALUATION CELLS

The envelope is fitted on aggregate cells rather than on individual windows. A cell is one (split, SIR stratum, overlap quartile) triple that retains at least 150 windows and all ten modulation classes. Overlap quartile edges are computed within each split. Table I gives the resulting construction; the fit total of 32 and the primary held total of 80 are both reconstructible from its rows.

Two features of Table I are worth stating plainly. First, the `id_test` split contributes 16 cells rather than 20 because its lowest overlap quartile is populated almost entirely by jammer-free windows, which carry $o = 0$ and the $\text{SIR} = 0$ dB cache sentinel; that bin clears the 150-window floor only in the 0 dB stratum. Second, the single `clean_retention` cell is excluded from the primary envelope because a jammer-free window has no physical finite SIR, which is also why its window count is the full split rather than a quartile of it.

III. ENVELOPE SKILL BY HELD REGIME

Table II reports the held-regime skill for both strong references, including two diagnostics that are easy to omit and easy to misread. “Const.” is the fit-domain constant predictor: the mean gain over the 32 fitting cells, applied unchanged to the held cells, with $R^2_{\text{skill}} = 1 - \text{MSE}_{\text{envelope}}/\text{MSE}_{\text{const}}$. “Oracle” is instead the mean of the held cells themselves (pooled row) or of the regime’s own held cells (regime rows), which no predictor could know in advance; it is reported as a hindsight reference, not as evidence that the envelope fails to resolve within-regime structure.

TABLE III

CLEAN-SENTINEL FITTING-CELL SENSITIVITY. THE 31-CELL FIT EXCLUDES THE SINGLE $\text{id_test/SIR} = 0/\text{LOWEST-OVERLAP}$ AGGREGATE DOMINATED BY JAMMER-FREE SENTINEL WINDOWS; THE HELD EVALUATION IS THE UNCHANGED 80 FINITE-SIR INTERFERED CELLS. RMSE VALUES IN PERCENTAGE POINTS; BE DENOTES THE BREAK-EVEN OVERLAP.

Reference	Fit cells	Held RMSE	Const. RMSE	R^2_{skill}	Overlap coeff.	SIR coeff.	BE o @ -15	BE o @ -10
IQFormer-insp.	32	6.23	8.37	0.45	14.38	-1.036	0.11	0.47
IQFormer-insp.	31	6.42	8.28	0.40	15.90	-1.066	0.14	0.47
MCLDNN	32	8.10	9.63	0.29	19.16	-0.894	0.06	0.29
MCLDNN	31	8.20	9.59	0.27	20.55	-0.922	0.08	0.31

The heterogeneity is real and is not smoothed away. Against IQFormer-inspired, the envelope transports the gain level well into the unseen-speed and held-out-channel regimes ($R^2_{\text{skill}} = 0.75$ and 0.59) and much less well into the unseen-jammer and combined-ODD regimes (0.20 and 0.34); against MCLDNN the unseen-jammer and combined-ODD regimes are at or below the constant baseline. The regime-wise oracle comparison is likewise heterogeneous: on the unseen-speed and held-out-channel shifts the envelope substantially beats the corresponding regime-specific constant (3.94 vs 6.71 and 4.86 vs 6.95 pp against IQFormer-inspired; 4.66 vs 7.78 and 5.41 vs 8.41 pp against MCLDNN), while on the unseen-jammer and combined-ODD shifts the regime-specific constant is better. The pooled oracle (6.04 and 6.81 pp) is lower than the pooled envelope RMSE because it uses the held-domain mean in hindsight; a pooled constant still bears the between-regime variation and the pooled comparison should not be read as the envelope lacking within-regime structure. The demonstrated property is therefore regime-dependent transport and within-regime structure, not uniform held-domain superiority.

IV. CLEAN-SENTINEL FITTING-CELL SENSITIVITY

The primary held envelope excludes jammer-free clean retention because a jammer-free window has no physical finite SIR. The fitting domain, however, retains one aggregate dominated by jammer-free sentinel windows, the $\text{id_test/SIR} = 0/\text{lowest-overlap}$ cell (mean overlap $\approx 10^{-11}$, 1053 windows, SIR taken from the cache sentinel). To close this definition asymmetry, we exclude exactly that aggregate, refit the two-variable envelope on the remaining 31 cells, and re-score the unchanged 80 finite-SIR interfered held cells; no model prediction changes and nothing is retrained.

The excluded aggregate is the $\text{id_test/SIR} = 0/\text{lowest-overlap}$ cell dominated by jammer-free sentinel windows. Excluding it moves the held RMSE by at most 0.20 percentage points and R^2_{skill} by at most 0.05, leaves both coefficient signs and the break-even ordering (required overlap rising as interference becomes less severe) unchanged, and therefore leaves the envelope conclusions materially unchanged; the 32-cell primary fit is retained.

V. CAPACITY CONTROL ACROSS ALL SPLITS

The capacity ladder isolates width from representation: S, M, and L differ only in three width numbers and share the routes, teacher, objective, and training recipe. All values in Tables IV and V are computed on the same matched five-seed subset, and must not be mixed numerically with ten-seed results reported elsewhere.

The pattern is consistent across splits. Width raises the average monotonically, and it helps most where the compact model was already competitive: at $\text{SIR} = -15$ dB the widest tier gains 4.96 percentage points over S and stays 14.07 percentage points ahead of IQFormer-inspired. It does not close the gap that matters for the boundary. On the marginal hard split L is still 2.84 percentage

TABLE IV

ABSOLUTE MACRO-F1 ON THE MATCHED FIVE-SEED SUBSET. THESE SPLIT-LEVEL FIVE-SEED MACRO-F1 VALUES SHOULD NOT BE NUMERICALLY DIFFERENCED AGAINST THE TEN-SEED CELL-MEAN GAINS IN THE MANUSCRIPT ENVELOPE TABLE.

Split	S	M	L	IQFormer-insp.	MCLDNN
hard interference	0.436	0.455	0.468	0.496	0.454
hard, SIR = −15 dB	0.353	0.384	0.403	0.262	0.255
in distribution	0.563	0.584	0.588	0.696	0.636
unseen jammer	0.456	0.462	0.466	0.612	0.584
unseen speed	0.487	0.507	0.510	0.626	0.582
held-out channel	0.488	0.510	0.515	0.613	0.567
combined OOD	0.450	0.462	0.465	0.621	0.587
clean retention	0.500	0.511	0.512	0.693	0.643

TABLE V

PAIRED CAPACITY CONTRASTS ON THE MATCHED FIVE-SEED SUBSET, IN PERCENTAGE POINTS WITH 95% INTERVALS.

Contrast (hard interference unless noted)	Diff.	Low	High
M minus S	1.94	1.07	2.80
L minus S	3.18	2.44	3.96
L minus IQFormer-insp.	−2.84	−4.30	−1.31
L minus MCLDNN	1.37	0.03	2.72
L minus S, SIR = −15 dB	4.96	3.48	6.48
L minus IQFormer-insp., SIR = −15 dB	14.07	9.55	18.51
L minus IQFormer-insp., clean retention	−18.14	−19.34	−17.00

points behind IQFormer-inspired, and on clean retention it is 18.14 percentage points behind, a deficit essentially untouched by tripling the width from M to L (0.10 percentage points of movement). This is the arithmetic behind the manuscript’s statement that capacity is not a substitute for the representation change.

VI. PROSPECTIVE TIER-2 INTERVENTIONS

The Tier-2 chain was designed after inspecting the frozen evidence and sits outside the sealed family. Table VI restates its three arms with the intervals that the compact form in the manuscript omits.

The ordering carries the argument. Adding jammer-free windows for every class moves clean macro-F1 by 0.15 percentage points, so exposure alone is not the explanation. Gating the modulation route on predicted interference presence moves nothing either, and the learned gate saturates near the modulated route in both clean and hard conditions (99.994% and 99.991%), so it functions as a negative control rather than as a demonstration that a mask fallback was learned. Only restoring a small amount of received-I/Q information moves both conditions with intervals clear of zero.

VII. SEED-LEVEL ROBUSTNESS

The sealed intervals treat algorithm seeds as fixed blocks. Treating them instead as a random effect and resampling seeds directly gives [4.02, 5.19] percentage points for the A5–A0 whole-configuration contrast, [−0.15, 0.42] percentage points for the teacher-form contrast, [−0.15, 1.04] percentage points for the route-count contrast, and [−0.19, 0.62] percentage points for the residual-free A7 variant against A5. The qualitative reading is unchanged under either treatment: the whole-configuration effect is resolved, the isolated component effects are not.

TABLE VI
PROSPECTIVE TIER-2 DIAGNOSIS, REPAIR, AND NEGATIVE CONTROL, RELATIVE TO A5, IN PERCENTAGE POINTS WITH 95% INTERVALS. EXPLORATORY THROUGHOUT.

Intervention	Params	Clean diff. [95% CI]	Hard diff. [95% CI]
Per-class clean coverage	39.5k	0.15 [−0.31, 0.63]	−0.30 [−0.91, 0.27]
Presence-gated route	39.5k	−0.30 [−0.66, 0.08]	−0.18 [−0.72, 0.35]
Received-I/Q sidecar	46.8k	11.77 [11.06, 12.49]	4.37 [3.44, 5.31]

TABLE VII
PER-SNR ACCURACY ON RML2016.10A OVER THREE SEEDS. BRACKETS ARE THE SEED MINIMUM AND MAXIMUM, NOT CONFIDENCE INTERVALS.

SNR (dB)	MCLDNN mean	[min, max]	IQFormer-insp. mean	[min, max]
−20	12.1%	[12.0, 12.2]	14.9%	[14.5, 15.2]
−18	13.1%	[12.3, 13.9]	18.1%	[16.8, 19.0]
−16	14.8%	[14.0, 15.3]	22.3%	[21.7, 23.0]
−14	17.9%	[16.8, 18.9]	27.2%	[26.6, 28.3]
−12	22.0%	[20.4, 23.3]	33.3%	[31.5, 34.6]
−10	26.0%	[25.0, 26.7]	37.0%	[36.5, 37.8]
−8	32.5%	[31.1, 33.3]	41.4%	[41.1, 41.5]
−6	44.0%	[42.8, 45.7]	49.8%	[49.5, 50.3]
−4	58.8%	[58.3, 59.9]	63.3%	[63.0, 63.5]
−2	74.7%	[74.3, 75.0]	77.3%	[76.1, 77.9]
0	83.4%	[82.9, 84.1]	86.1%	[85.7, 86.7]
2	85.1%	[84.7, 85.6]	89.4%	[88.6, 90.5]
4	85.5%	[84.7, 87.1]	91.6%	[90.6, 93.0]
6	86.1%	[85.7, 86.8]	93.4%	[93.0, 94.1]
8	86.4%	[84.9, 88.6]	93.3%	[92.8, 94.2]
10	85.3%	[84.0, 87.1]	93.0%	[92.0, 94.4]
12	86.3%	[85.1, 88.4]	93.8%	[93.1, 94.7]
14	86.0%	[85.0, 87.9]	93.4%	[92.9, 93.9]
16	85.5%	[84.5, 87.1]	92.9%	[92.1, 93.7]
18	85.2%	[84.3, 86.8]	92.9%	[92.3, 93.9]
All SNR	58.55%	[58.1, 59.4]	65.23%	[64.8, 65.8]
Reference release	62.05%	—	64.19%	—

VIII. COMPARATOR FIDELITY ON A PUBLIC BENCHMARK

This check exists to establish that the two I/Q comparators are implemented competently, not to validate the simulator-domain envelope; a public benchmark cannot do the latter, because it does not expose the separately tracked target and jammer components the overlap covariate requires. The protocol follows the benchmark result table released with the IQFormer reference implementation: the 11-class RML2016.10a set, the full −20 to 18 dB grid, per-stratum 600/200/200 splits seeded at 233, three algorithm seeds, no augmentation and no input-length change. The public-benchmark fidelity runs use a separate recipe following the benchmark protocol—a 60-epoch budget for IQFormer-inspired and a 200-epoch budget for MCLDNN—and should not be used to infer convergence sufficiency of the 30-epoch main campaign.

Table VII makes the shape of the agreement explicit. Both implementations reproduce the reference release at the level of the all-SNR range, and IQFormer-inspired matches it closely (+1.04 percentage points), while MCLDNN remains 3.50 percentage points lower but within the same broad range. Agreement is not pointwise: individual SNR bins deviate by up to about 8 percentage points, which

TABLE VIII
CONVERGENCE AUDIT OVER THE TEN SEALED TRAINING HISTORIES PER MODEL. “AT EPOCH 30” COUNTS SEEDS WHOSE TRAINING RAN THE FULL SHARED BUDGET; “EARLY-STOPPED” COUNTS SEEDS STOPPED EARLIER BY THE PATIENCE RULE.

Model	Median best epoch	Range	Seeds at epoch 30	Early-stopped seeds
MCLDNN	13	11–16	0/10	10/10
IQFormer-insp.	27	24–30	10/10	0/10
CSSL	29	26–30	10/10	0/10
A5	29.5	27–30	10/10	0/10
A0	30	28–30	10/10	0/10

is why this is reported as range-level implementation fidelity rather than reproduction. No fidelity threshold was declared before these runs, so no pass/fail band is claimed here.

IX. COMPARATOR CONVERGENCE AUDIT

All models were trained under one shared 30-epoch budget with the same cosine-scheduled learning rate and an 8-epoch early-stopping rule; the reported checkpoint is selected solely by validation modulation cross-entropy. Table VIII audits the sealed training histories (selected epoch and completed epochs over the ten algorithm seeds) to check whether any comparator’s comparison is distorted by an immediately binding training cap.

The budget binds asymmetrically. IQFormer-inspired and CSSL run the full 30-epoch budget in all ten seeds (median selected epoch 27 and 29), so their reported performance should be interpreted as potentially budget-limited under this shared 30-epoch protocol and the shared epoch budget may favor faster-converging models; the manuscript states this caveat in its Limitations. MCLDNN early-stops in every seed with best epochs well inside the budget (median 13, range 11–16), so the severe-region comparisons against MCLDNN are not explained by a binding cap. A5 itself also trains to the budget in all ten seeds, and the direct spectral reference A0 likewise runs to the budget in all ten seeds (median selected epoch 30, range 28–30), so the cap concern is not one-sided against the comparators. No model was retrained for this audit; the table reads the sealed histories only.

X. SIDECAR ARCHITECTURE DETAIL

Table IX lists the received-I/Q sidecar layer by layer so the reported parameter increment is fully reconstructible. All convolution layers are bias-free; SiLU activations (parameter-free) follow each normalization. The branch consumes the received two-channel mixture at the full sample rate, uses one depthwise-separable stack with aggressive time pooling (mean/std/max over the window), and fuses the resulting embedding residually with the spectral embedding before a side classification head. The fused embedding passes through the 10-class linear head of layer 13, and these logits are added residually to the base spectral classifier logits; the side head receives no separate auxiliary supervision, and inference consumes only the received mixture. The parameter counts were verified directly against the sealed checkpoint state dict.

XI. ENVELOPE COVARIATE LADDER AND SELECTOR AUDIT

Two zero-compute audits recompute, on the same 32 fitting cells and the same 80 finite-SIR interfered held cells as the primary envelope, (i) a covariate ladder that isolates the skill of each physical covariate and (ii) a model-selection selector evaluation. No model is retrained and no prediction changes; both are exploratory.

TABLE IX

SIDECAR LAYER STRUCTURE AND PARAMETER COUNT. ALL CONV1D LAYERS USE STRIDE 1 AND NO BIAS; DW DENOTES DEPTHWISE (GROUPS = IN CHANNELS) AND PW POINTWISE (1×1) CONVOLUTION. TOTAL MODEL SIZE IS $39,500 + 7,294 = 46,794$ PARAMETERS.

Layer	Type	Kernel	Stride	In→Out ch.	Params
1	Conv1D	9	1	2→12	216
2	GroupNorm (3 groups)	—	—	12	24
3	DW Conv1D	7	1	12→12	84
4	PW Conv1D	1	1	12→24	288
5	GroupNorm (6 groups)	—	—	24	48
6	DW Conv1D	5	1	24→24	120
7	PW Conv1D	1	1	24→24	576
8	GroupNorm (6 groups)	—	—	24	48
9	mean/std/max pool + Linear	—	—	72→24	1,752
10	LayerNorm	—	—	24	48
11	Fusion Linear	—	—	72→48	3,504
12	Fusion LayerNorm	—	—	48	96
13	Classifier Linear	—	—	48→10	490
Total sidecar increment					7,294

TABLE X

COVARIATE LADDER ON THE SHARED 32 FITTING / 80 HELD CELLS. RMSE IN PERCENTAGE POINTS. INCREMENTS: OVERLAP|SIR REDUCES HELD RMSE BY 0.61 PERCENTAGE POINTS AGAINST BOTH REFERENCES; SIR|OVERLAP BY 1.23 PERCENTAGE POINTS AGAINST IQFORMER-INSPIRED AND 0.58 PERCENTAGE POINTS AGAINST MCLDNN.

Model	IQ RMSE	IQ R^2_{skill}	MC RMSE	MC R^2_{skill}
Constant	8.37	0.00	9.63	0.00
SIR only	6.84	0.33	8.71	0.18
Overlap only	7.46	0.20	8.68	0.19
Overlap + SIR	6.23	0.45	8.10	0.29
Overlap + SIR + SNR	6.15	0.46	8.05	0.30

Table X gives the ladder. All models are fitted on the 32 fitting cells by least squares and scored unchanged on the 80 held cells; R^2_{skill} uses the same fit-domain constant baseline as the manuscript. SIR is the stronger single covariate against IQFormer-inspired, whereas SIR and overlap have comparable single-covariate skill against MCLDNN; overlap adds a consistent increment beyond SIR against both references, while SIR adds more beyond overlap against IQFormer-inspired than the reverse. The statistical status of that increment is assessed by the paired bootstrap in Section XII.

Table XI evaluates the envelope as a model-selection rule on the 80 held cells. Each sign selector uses envelope coefficients fitted only on the 32 fitting cells and applied to held cells by covariate value; the oracle is a hindsight per-cell best and is not achievable. Realized score is the mean over cells of the selected model’s actual macro-F1; MACs use the sealed complexity table values (41.8 M for A5, 355.6 M for IQFormer-inspired).

The reading is deliberately plain. Inside the held finite-SIR regimes the large I/Q reference leads in almost every cell, so no selector recovers much against always using it, and the envelope’s demonstrated value is boundary localization—identifying the few severe, high-overlap cells where the compact model is competitive—rather than a large expected compute saving in this campaign. The overlap+SIR rule is the only rule that ever selects the compact model (5 of 80 cells), at a regret of 0.36 percentage points against the hindsight oracle; the SIR-only rule never selects it. Computed

TABLE XI

SELECTOR EVALUATION ON THE 80 HELD CELLS. REGRET IS AGAINST THE HINDSIGHT ORACLE; EXPECTED MACS AVERAGE THE SELECTED MODEL’S COST OVER CELLS. THESE SELECTOR STATISTICS ARE COMPUTED FROM THE TEN-SEED HELD-CELL AGGREGATES AND SHOULD NOT BE NUMERICALLY DIFFERENCED AGAINST THE MATCHED-FIVE-SEED SPLIT-LEVEL MACRO-F1 VALUES IN TABLE IV.

Selector	Realized F1	Regret (pp)	A5 cells	MACs (M)	MAC saving
Always A5	0.4488	14.19	80/80	41.8	88.2%
Always IQFormer	0.5894	0.14	0/80	355.6	0%
SIR-only sign rule	0.5894	0.14	0/80	355.6	0%
Overlap+SIR sign rule	0.5872	0.36	5/80	336.0	5.5%
Oracle (hindsight)	0.5907	0	3/80	343.8	3.3%

TABLE XII

LEAVE-15-DB-OUT TEST. ALL PREDICTORS ARE FITTED ON THE 28 FITTING CELLS EXCLUDING THE FOUR `HARD_INTERFERENCE` SIR = -15 dB CELLS AND SCORED ON THOSE FOUR CELLS ONLY. RMSE AND MAE IN PERCENTAGE POINTS; SIGN ACCURACY IS THE FRACTION OF THE FOUR CELLS WHOSE PREDICTED GAIN SIGN MATCHES THE OBSERVED SIGN.

Reference	Predictor	RMSE	MAE	Sign acc.
IQFormer-insp.	Constant	21.34	21.26	0/4
IQFormer-insp.	SIR only	12.15	12.02	0/4
IQFormer-insp.	Overlap + SIR	12.69	12.01	1/4
MCLDNN	Constant	16.43	15.83	0/4
MCLDNN	SIR only	7.87	6.54	4/4
MCLDNN	Overlap + SIR	6.84	6.52	2/4

directly from the held-cell artifacts, its confusion matrix against the hindsight-optimal indicator is TP= 3, FP= 2, FN= 0, TN= 75: it identifies all three compact-favoring cells (recall 3/3) with two additional selections (precision 3/5), so its held-domain value is boundary localization without missed positive cells, while always using IQFormer-inspired still attains the higher average realized macro-F1. This is the honest selector consequence of a regime structure in which the compact model’s advantage is concentrated in a small corner of the plane.

XII. LEAVE-15-DB-OUT EXTRAPOLATION AND PAIRED OVERLAP-INCREMENT TESTS

Two further zero-compute audits close the remaining envelope questions. Both reuse the frozen cell table; no model is retrained and no prediction changes.

The 80 held cells contain no SIR = -15 dB cell: that stratum occurs only in the `hard_interference` fitting split. To probe extrapolation toward the most severe level, we refit the two-variable envelope on the 28 fitting cells excluding the four -15-dB cells and predict those four cells out-of-sample (a leave-one-SIR-stratum-out test, not an external validation). Table XII compares the envelope with a 28-cell constant and a 28-cell SIR-only fit.

The outcome is reference-dependent. Against MCLDNN the 28-cell envelope preserves the severe-region rank reversal: all four held-out cells favor A5 in hindsight, and the envelope’s 6.84 percentage points RMSE is markedly better than the 16.43 percentage points constant and the 7.87 percentage points SIR-only fit. Against IQFormer-inspired the direction is not preserved: the fitted surface underpredicts the strongly positive -15-dB gains (three of four sign errors), and the full model is no better than SIR-only. The -15-dB break-even overlaps are therefore conditional extrapolations of the fitted

TABLE XIII

BLOCK-PAIRED BOOTSTRAP OF THE OVERLAP INCREMENT BEYOND SIR ON THE 80 HELD CELLS. ΔMSE IS THE PAIRED SQUARED-ERROR REDUCTION (SIR-ONLY MINUS OVERLAP+SIR), IN pp^2 ; INTERVALS ARE 2.5/97.5 PERCENTILES OVER 10,000 BLOCK RESAMPLES.

Reference	Mean ΔMSE	95% CI	$P(\Delta\text{MSE} > 0)$	RMSE red. (pp)
IQFormer-insp.	7.97	$[-2.06, 18.18]$	0.94	0.61
MCLDNN	10.18	$[-8.29, 29.84]$	0.85	0.61

envelope—well supported against MCLDNN, not validated against IQFormer-inspired—rather than independently held-regime-validated thresholds.

For the incremental value of overlap beyond SIR on the 80 held cells, we use a block-paired bootstrap. The paired statistic is the per-cell squared-error difference $d_i = e_{\text{SIR-only},i}^2 - e_{\text{overlap+SIR},i}^2$, so $\bar{d} > 0$ means adding overlap lowers MSE. The 80 held cells form 20 held-regime $\times$ SIR blocks of four overlap quartiles each; each draw resamples the 20 blocks with replacement ($B = 10,000$, seed 20260820) and keeps all quartiles inside a sampled block.

Both references show positive point estimates and a majority of positive draws, but both percentile intervals include zero; overlap therefore provides a consistent point-estimate refinement beyond SIR rather than a separately resolved predictor, and the manuscript reports it as an exploratory refinement.

Finally, applying the 28-cell sign rule to the four excluded -15-dB cells with the same ten-seed cell-level aggregates and MAC cost model selects A5 in only one of the four hindsight-A5-favoring cells (sign accuracy 1/4; realized cell-mean macro-F1 0.285 versus the 0.361 hindsight oracle on those four cells). The cross-model selector therefore transfers poorly to the excluded severe stratum, which is the selector-side counterpart of the extrapolation caveat above.

XIII. INDEPENDENT SEVERE-HELD VALIDATION: TRANSFER FAILURE OF THE CORNER ADVANTAGE AND THE ENVELOPE

After the campaign was frozen, a preregistered post-hoc robustness validation asked whether the severe-corner advantage and the overlap-SIR envelope survive on data generated outside the frozen campaign. A new severe held set was simulated with fresh source IDs that are disjoint from the campaign source pool (12,820 new sources against the campaign’s 152,000; the disjointness is asserted programmatically at cache build), at the two severe regimes and view 0, and stored under `standards/cache_s5_severe_held_v1/` with a manifest recording the cache digest. All frozen checkpoints were evaluated unchanged; no model was retrained, tuned, or selected against this set. The validation is exploratory and post-hoc relative to the sealed family throughout.

Table XIV reports the preregistered transfer verdicts. The pooled A5 gain is negative against both comparators, the envelope’s sign accuracy is near chance (and below it against MCLDNN), a SIR-only or constant predictor beats the full overlap-SIR envelope on RMSE against both references, and the overlap-SIR cell selector recalls only 0.581 of the truly A5-favoring cells. The preregistered verdict is therefore a transfer failure: neither the severe-corner advantage nor the -15-dB envelope transport reproduces on independently generated severe data.

XIV. INDEPENDENT SEVERE-HELD VALIDATION: REPRESENTATION REPAIR BY THE I/Q SIDECAR

A second preregistration, frozen before any inference on the new held set, then asked whether the received-I/Q sidecar repairs the failure exposed above. The data stay locked to the same cache; the A5,

TABLE XIV

INDEPENDENT SEVERE-HELD TRANSFER CHECK (POST-HOC, EXPLORATORY). POOLED GAIN IS A5 MINUS THE REFERENCE IN MACRO-F1 PERCENTAGE POINTS; RMSE ROWS COMPARE THE REFITTED ENVELOPE AGAINST SIR-ONLY AND CONSTANT PREDICTORS, IN PERCENTAGE POINTS.

Reference	Pooled gain	Sign acc.	RMSE full	RMSE SIR-only	RMSE const.
IQFormer-insp.	-3.12	0.578	9.75	9.05	8.11
MCLDNN	-12.45	0.406	22.56	22.29	15.41

IQFormer-inspired, and MCLDNN predictions are reused from the transfer check; and the ten sidecar checkpoints pass a provenance audit (46794 parameters each, strict loading of the frozen state dicts, SHA-256 recorded in `docs/S5_R1_CHECKPOINT_AUDIT.md`). The primary contrast is sidecar minus A5, pooled as the equally weighted mean of the per-seed paired differences, with a seed-level paired bootstrap interval.

The repair is robust. Pooled macro-F1 rises from 20.72% (A5) to 25.09% (sidecar), a difference of 4.37 percentage points ([2.96, 5.90]), with regime contributions 4.22 and 4.52 percentage points; the difference is positive in 10 of 10 seeds and in 8 of eight SNR strata. A window-level hierarchical paired audit reproduces the interval. On the same held set the sidecar also overtakes IQFormer-inspired (23.29%; descriptive gap-recovery ratio 1.70) while recovering only 0.34 of the remaining gap to MCLDNN (33.75%). The preregistered verdict is therefore a strong repair effect: the representation repair generalizes beyond the campaign even where the campaign-internal corner advantage and the envelope transport do not. Per-seed, per-SNR, and per-class tables are stored as CSV (artifact index below); the sidecar remains exploratory and outside the sealed family.

XV. ARTIFACT INDEX

Every number in this document is derived from the frozen artifact tree rather than transcribed from an earlier draft. The relevant entry points are:

- Cell construction and envelope fits: `analysis_zero_compute/outputs/csv/a14_envelope_cells.csv` and `a14_v41_envelope_interfered_holdout.csv`. The V4.4 clean-sentinel 31-cell sensitivity is recomputed from the same cell table by `analysis_zero_compute/a16_clean_sentinel_sensitivity.py`, writing `a16_clean_sentinel_sensitivity.csv` and `a16_clean_sentinel_sensitivity.json`.
- Covariate ladder, selector evaluation, and convergence audit: `analysis_zero_compute/a17_final_s2_audits.py`, writing `a17_covariate_ladder.csv`, `a17_selector_evaluation.csv`, and `a17_convergence_audit.csv` from the same frozen cell table and the sealed per-seed `result.json` training histories.
- Leave-15-dB-out test, paired overlap-increment bootstrap, and selector confusion matrix: `analysis_zero_compute/a18_final_s4_audits.py`, writing `a18_leave15_out.csv`, `a18_overlap_increment_bootstrap.csv`, `a18_selector_confusion.csv`, and `a18_leave15out_selector.csv` from the same frozen cell table and the sealed per-seed predictions.
- Sidecar architecture: `analysis_zero_compute/tier2_gpu/run_tier2_experiment.py` (LightweightIQSidecarVIMD); the parameter counts in Table IX are verified against the sealed checkpoint state dict under `artifacts/tier2_iq_sidecar_v1/models/`.

- Capacity ladder: tier2_capacity_L_v1_vs_S.csv, tier2_capacity_L_v1_vs_M.csv, tier2_capacity_L_v1_vs_iqformer.csv, tier2_capacity_L_v1_vs_mclnncnn.csv, and tier2_capacity_M_v2_vs_a5.csv, all five-seed matched.
- Tier-2 interventions: tier2_iq_sidecar_v1_vs_a5.csv, tier2_condition_coverage_v1_vs_a0.csv, and tier2_presence_gated_v1_vs_a5.csv.
- Comparator fidelity: artifacts/rml2016_10a_fidelity_v3/, whose run.json records the dataset SHA-256, the split protocol, the reference-release values, and the environment.
- The overlap covariate is computed at cache-build time, stored as overlap.npy per split, and SHA-256 checksummed in the sealed cache manifest; it is never an input feature of any model.
- Independent severe held cache: standards/cache_s5_severe_held_v1/; the manifest records the cache digest and the programmatic source-disjointness assertion against the campaign source pool.
- Severe-held transfer check (Table XIV): analysis_zero_compute/a19_s5_severe_held.py, writing analysis_zero_compute/outputs/a19_s5_severe_held.json together with the per-cell CSV files, from the frozen predictions under artifacts/s5_severe_held_v1/.
- Severe-held repair check: analysis_zero_compute/a20_s5_r1_sidecar_severe_held.py, writing the sidecar predictions under artifacts/s5_r1_sidecar_severe_held_v1/ and a20_s5_r1_model_levels.csv, a20_s5_r1_paired_contrasts.csv, a20_s5_r1_per_snr.csv, a20_s5_r1_per_class.csv, a20_s5_r1_gap_recovery.csv, and a20_s5_r1_summary.json; the checkpoint provenance audit is docs/S5_R1_CHECKPOINT_AUDIT.md.

The reproduction package released with the paper contains the resolved macros, the figure-source CSV files, per-seed aggregates, run configurations, and the full provenance manifest.